

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	27/8/2026	Duration:	60 minutes

Code of Conduct

I G. Akshya (192425132) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Akshya

Co4 Assessment Tool 1

Name: -G. Akshya

Reg: -192425132

Course Code: -CSA1007

Slot: -B

1. Introduction to the Emerging Technology

Smart Vertical Farming Automation is an emerging application of Internet of Things (IoT), automation, data analytics, and artificial intelligence for managing crops grown in vertically stacked indoor environments. Instead of depending completely on manual observation, the platform continuously measures environmental conditions and uses the collected information to support or automate decisions related to irrigation, lighting, ventilation, and crop monitoring.

Problem Statement

Traditional crop monitoring can require frequent manual checking and may not respond quickly to changing environmental conditions. The proposed platform addresses this by combining continuous monitoring with automated control. The system is intended to make irrigation and lighting decisions more systematic while giving operators timely information about farm conditions and possible crop-health issues.

2. Need and Motivation

The need for a smart vertical farming platform arises from the requirement to produce crops in a controlled environment while using limited resources efficiently. Vertical farming allows crops to be grown in stacked layers, but the controlled environment also requires continuous management of water, lighting, temperature, humidity, and other conditions.

- Continuous monitoring: sensor readings can provide frequent information about the growing environment.
- Efficient irrigation: watering can be based on moisture conditions instead of relying only on fixed schedules.
- Energy-aware lighting: automated lighting can maintain a suitable light cycle while avoiding unnecessary operation.
- Better decision-making: dashboards and reports can help operators understand production and resource usage.

3. Working Principle / Architecture

The platform follows a sensing–processing–decision–actuation cycle. Sensors first collect environmental data. An IoT controller or gateway receives the readings and sends them to the local decision logic and, where required, a cloud database. The decision engine evaluates the readings and determines whether an action is required. Actuators then control irrigation, lighting, fans, or other farm equipment. The resulting data can be displayed through a dashboard and used to generate reports and alerts.

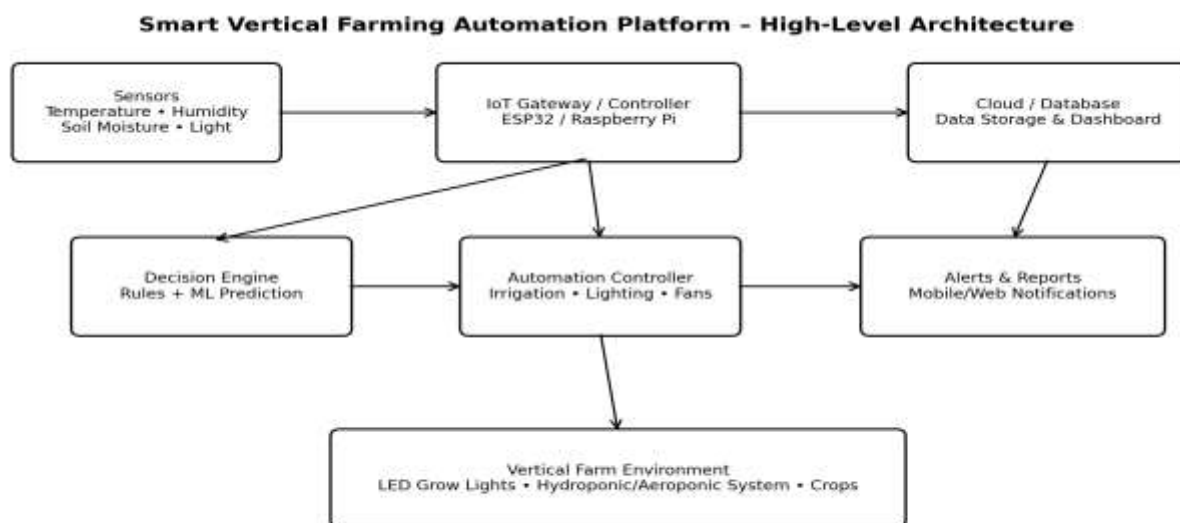


Figure 1. High-level architecture of the Smart Vertical Farming Automation Platform

System Workflow

Step	Stage	Description
1	Sense	Sensors collect temperature, humidity, moisture and light readings.
2	Transmit	The IoT controller receives and forwards the sensor data.
3	Analyse	Rules or prediction models evaluate the current and historical data.
4	Decide	The system determines whether irrigation, lighting or another action is needed.
5	Actuate	Control devices are switched or adjusted automatically.
6	Monitor	The dashboard displays status and stores data for reports and future analysis.
7	Alert	Operators are notified when readings or predicted conditions require attention.

4. Key Components and Technologies

Component	Example Technology	Purpose
Environmental Sensors	Temperature, humidity, moisture and light sensors	Collect real-time farm conditions.
IoT Controller	ESP32, Arduino-class controller or Raspberry Pi	Reads sensors and controls connected devices.
Actuators	Water pump, valves, LED lights, fans	Perform automated farm actions.

Communication	Wi-Fi / local network / IoT protocol	Transfers sensor and control information.
Database	Cloud or local database	Stores historical readings, events and reports.
Dashboard	Web or mobile interface	Shows current conditions, trends and alerts.
Decision Engine	Threshold rules and/or ML models	Converts sensor data into control decisions.
Notification Module	Email, SMS or app notifications	Notifies operators about important events.

Software Engineering Perspective

From a software engineering perspective, the platform can be organized into independent modules so that monitoring, prediction, control, reporting and notification functions can be developed and tested separately. This modular design supports maintainability and allows individual components to be upgraded without redesigning the complete system.

Examples of Automated Actions

- ❖ If moisture falls below a configured threshold, the irrigation system can be activated for a controlled period.
- ❖ If light intensity or the programmed lighting cycle is insufficient, LED grow lights can be switched on.
- ❖ If temperature or humidity moves outside the desired range, the system can generate an operator alert.
- ❖ If repeated sensor patterns indicate possible plant stress, the prediction module can flag the crop area for inspection.
- ❖ Daily or weekly sensor information can be summarized into production and resource-use reports.

6. Advantages and Limitations

Advantages

- Higher potential crop yield through better control of growing conditions.
- Reduced water consumption through condition-based irrigation.
- Improved resource efficiency by coordinating irrigation, lighting and monitoring.
- Better crop quality through consistent environmental management.

Limitations

- ❖ Initial installation cost for sensors, controllers, actuators and networking equipment.
- ❖ Sensors require calibration, maintenance and occasional replacement.
- ❖ Network or power failures can affect remote monitoring and automation.
- ❖ Prediction accuracy depends on the quality and quantity of collected data.

7. Recent Developments and Trends

Smart farming is increasingly moving from simple sensor monitoring toward integrated systems that combine IoT data, automation, analytics and AI. For vertical farming, this trend supports more precise control of the crop environment and better use of historical data.

- AI-assisted crop-health prediction using environmental and image data.
- Computer vision for detecting visible plant stress, disease symptoms and growth patterns.
- Edge computing for faster local decisions when cloud connectivity is limited.

8. Challenges and Future Scope

Challenges

- Reliable sensor readings are essential because incorrect measurements can lead to incorrect control actions.
- The system must remain safe when communication or hardware components fail.
- Automation decisions need suitable thresholds, rules or trained models.
- Large farms may produce high volumes of sensor data that require organized storage and analysis.

Future Scope

- Develop computer-vision models that automatically inspect plant images.
- Use machine learning to predict crop health and optimize irrigation schedules.
- Add adaptive lighting that responds to crop growth stage and environmental conditions.
- Integrate renewable-energy information to improve sustainability decisions.
- Add digital-twin or simulation capabilities for testing farm-control strategies before deployment.

Illustrative Input–Decision–Output Table

Input Condition	Decision	Output
Low moisture	Start irrigation	Pump/valve activated
Adequate moisture	No irrigation	Pump remains off

Low light during required period	Enable lighting	LED lights activated
High temperature	Check ventilation	Fan/alert action
Abnormal sensor trend	Flag for inspection	Operator notification

10. Conclusion

The Smart Vertical Farming Automation Platform provides a structured approach to managing crops in a controlled vertical farming environment. By combining environmental sensing, IoT control, automation, data storage, reporting and intelligent prediction, the platform can reduce dependence on continuous manual monitoring.

The objectives provided in the problem statement are directly addressed through environmental monitoring, optimized irrigation, crop-health prediction, automated lighting, production reporting and operator notifications.

11. Objectives

1. Monitor environmental conditions.
2. Optimize irrigation.
3. Predict crop health.
4. Automate lighting control.
5. Generate production reports.
6. Notify farm operators.